Multimedia Signal Processing - 2nd module

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1 Question - 10 pts

Consider a gray-scale image with 3 grey levels $L = 0, 1, 2$. The following statistics can be estimated collecting data from an image collection:

$$p(0) = \Pr(l = 0) = 0.25$$
$$p(1) = \Pr(l = 1) = 0.35$$
$$p(2) = \Pr(l = 2) = 0.40$$

- Compute the entropy of the source.
- Design a Huffman code using block sizes $N = 1, 2$.
- In each case, compute the average code length L_N.

Inter-pixel correlation is described by the following statistics

$$P = \begin{bmatrix} 0.10 & 0.10 & 0.05 \\ 0.05 & 0.25 & 0.05 \\ 0.10 & 0.00 & 0.30 \end{bmatrix} \quad (1)$$

where p_{ij} denotes the probability $\Pr(l_1 = i, l_2 = j)$, given two consecutive pixels l_1 and l_2 in raster scan order.

- Show that pixel values of consecutive pixels are statistically dependent
- Compute the joint entropy
- Show how to compute the conditional probabilities that need to be used in an arithmetic coder. Explicitly compute $\Pr(l_1 = 0|l_2 = 0)$.

2 Question - 5 pts

Illustrate the block diagram of a lossy video coding architecture based on transform coding (both encoder and decoder).

3 Questions - 5 pts (each answer can be either TRUE or FALSE)

1. Let X denote a discrete memoryless source with finite alphabet, and $D(R)$ the corresponding rate-distortion function, σ_x^2 its variance and $H(X)$ its entropy.
   - [T] [F] $H(X) < \infty$
   - [T] [F] $D(0) = \sigma_x^2$
   - [T] [F] $R(0) \geq H(X)$
2. Consider a perceptual audio coder
   - [T] [F] The analysis filter bank is the same used by the perceptual model
   - [T] [F] Noise maskers have higher SMR than tonal maskers.
   - [T] [F] All critical bands have the same bandwidth.
   - [T] [F] Noise and tonal maskers are lossless coded
3. Consider a lossy image coding system
   - [T] [F] The computational complexity of encoding is significantly higher than decoding.
   - [T] [F] Coding efficiency increases when the block size increases.
   - [T] [F] The DCT transform achieves perfect reconstruction (up to arithmetic precision)

4 Question - 12 pts

Local image features are routinely used in several computer vision applications, including augmented reality (e.g Google Goggles). Their success is due to ability to summarize the input image in a way that is invariant to several image transformations (scale, rotation, luminance, viewpoint etc...).

- Load the file `feature_set.mat`. It contains a collection of 128-dimensional SIFT local features extracted from a test image. Each dimension of each feature vector is quantized with 8 bits.
- Assume the feature set as a memoryless source that produces symbols from 0:255. What is the minimum number of bits to lossless encode the feature set?

Recent studies have shown that SIFT feature vectors are robust to heavy quantization, i.e. the accuracy of object recognition does not decrease when local features are lossy coded.

- Design a uniform quantizer with 2 reconstruction levels and compute the resulting MSE.
- Design an optimal quantizer with 2 reconstruction levels using the Lloyd-Max algorithm. Plot the resulting MSE for each iteration of the algorithm.